

Part A

Question 1:

Let

$$A = \{1, 2, 3, 4, 5\}, \quad B = \{cat, dog, rabbit, hamster\}, \quad C = \mathbb{Z}, \quad D = \{\alpha, \beta, \gamma, \delta\}$$

a) You've mislabeled the question parts, e.g. (i) was "bijection"

i.

$$f : A \rightarrow B,$$

$$f(1) = cat, f(2) = hamster, f(3) = dog, f(4) = rabbit, f(5) = cat$$

f is onto but not one-to-one.

ii.

$$g : A \rightarrow D,$$

$$g(1) = \alpha, g(2) = \delta, g(3) = \beta, g(4) = \beta, g(5) = \alpha$$

g is neither one-to-one nor onto

iii.

$$F : A \rightarrow C,$$

$$F(a) = 100a, \text{ for all } a \in A$$

$F(a) = 100a$ is one-to-one but not onto

iv.

$$\phi : B \rightarrow D,$$

$$\phi(cat) = \alpha, \phi(dog) = \gamma, \phi(rabbit) = \beta, \phi(hamster) = \delta$$

ϕ is a bijection.

b)**i.**Example of a map $S \rightarrow T$ that is not one-to-one:

$$2 \mapsto 1, 3 \mapsto 1, 5 \mapsto 5, 7 \mapsto 7$$

.

ii.Example of a bijection $S \rightarrow T$:

$$2 \mapsto 1, 3 \mapsto 3, 5 \mapsto 5, 7 \mapsto 7$$

.

3/3

Question 2:

for $n \geq 1$, $7^{2n} - 6^n$ is divisible by 43

Proof.

Using proof by induction we will first prove the basis of induction $n = 1$

$$\begin{aligned} \times \quad P(1) &= 7^{2(1)} - 6^1 \\ &= 49 - 6 \\ &= 43 \end{aligned}$$

You should define $P(n)$ to be a statement, not a number. Specifically let $P(n)$ stand for " $7^{2n} - 6^n$ is divisible by 43".

Hence $P(1)$ is divisible by 43

Now we will assume that $P(k)$ is true for some arbitrary $k \geq 1$

$$\begin{aligned} \times \quad P(k) &= 7^{2k} - 6^k \\ &= 43m \end{aligned}$$

for some integer m

Next we will use this as our inductive hypothesis to prove that $P(k+1)$ is true

$$\begin{aligned} \times \quad P(k+1) &= 7^{2(k+1)} - 6^{k+1} \\ &= 7^{2k+2} - 6^{k+1} \\ &= 7^2 \cdot 7^{2k} - 6 \cdot 6^k \\ &= 49 \cdot 7^{2k} - 6 \cdot 6^k \end{aligned}$$

If you want to introduce a notation for $7^{2n} - 6^n$, call it e.g. $f(n)$, and make sure to still introduce the statement $P(n)$ (which would now be " $P(n): 43|f(n)$ ").

From our inductive hypothesis we know that $7^{2k} - 6^k = 43m$

$$\begin{aligned} &= 49(43m + 6^k) - 6 \cdot 6^k \\ &= 49 \cdot 43m + 49 \cdot 6^k - 6 \cdot 6^k \end{aligned}$$

Factorising out the 6^k

$$= 49 \cdot 43m + (49 - 6) \cdot 6^k$$

$$= 49 \cdot 43m + 43 \cdot 6^k$$

Factoring out the 43

$$= 43(49m + 6^k)$$

Since m and k are both integers, $49m + 6^k$ is also an integer

hence $P(\overbrace{k+1}^{2(k+1)-k})$ is divisible by 43

good

3/4

By the principle of mathematical induction we have shown that $P(n)$ is true for all $n \geq 1$

□

Question 3:

$$\text{hcf}(2025, 630) = 2025x + 630y$$

$$2025 = 3 \cdot 630 + 135$$

$$630 = 4 \cdot 135 + 90$$

$$135 = 1 \cdot 90 + 45$$

$$90 = 2 \cdot 45 + 0$$

Hence,

$$\text{hcf}(2025, 630) = 45$$

rearranging to make the remainder the subject;

$$45 = 135 - 1 \cdot 90$$

$$90 = 630 - 4 \cdot 135$$

$$135 = 2025 - 3 \cdot 630$$

Substituting back up the chain we get;

$$45 = 135 - 1(630 - 4 \cdot 135)$$

$$= 5 \cdot 135 - 1 \cdot 630$$

$$= 5(2025 - 3 \cdot 630) - 1 \cdot 630$$

$$= 5 \cdot 2025 - 15 \cdot 630 - 1 \cdot 630$$

$$= 5 \cdot 2025 - 16 \cdot 630$$

Hence one solution is

$$x = 5, y = -16$$

4/4

Question 4:**a)**

If a and b are positive integers and p divides $\text{hcf}(a^2, b)$ if, and only if, p divides $\text{hcf}(a, b^2)$.

Proof.

Let p be a prime number.

Forward direction ($\Rightarrow$): Suppose $p \mid \text{hcf}(a^2, b)$ then by Definition 4.5, HB p15,

$$p \mid a^2 \text{ and } p \mid b$$

Since p is a prime number, and $p \mid a^2$, by Euclid's Lemma for prime factors (Theorem 1.4, Chapter 2 HB p17),

$$p \mid a$$

Hence,

$$p \mid a \text{ and } p \mid b$$

Which implies, by Definition 4.5, HB p15,

$$p \mid \text{hcf}(a, b^2) \because p \mid b \implies p \mid b^2$$

Backward direction ($\Leftarrow$): conversely suppose $p \mid \text{hcf}(a, b^2)$ Then

$$p \mid a \text{ and } p \mid b^2$$

Again by Theorem 1.4, HB p17,

$$p \mid b^2 \implies p \mid b$$

Therefore,

$$p \mid a \text{ and } p \mid b \implies p \mid a^2 \text{ also}$$

Hence $p \mid (a^2, b)$

Thus, we have shown in both directions and so;

$$p \mid \text{hcf}(a^2, b) \iff p \mid \text{hcf}(a, b^2)$$

□

Definition 4.5, Chapter 1 HB p15: Highest common factor(HCF), $\text{hcf}(a, b)$, of two integers a and b , not both of which are zero, is the natural number n satisfying

$$(a)n \mid a \text{ and } n \mid b;$$

$$\text{if } d \mid a \text{ and } d \mid b \text{ then } d \leq n$$

great!
3/3

b)

If a and b are positive integers and m and n are defined by

$$m = 3a + b \quad \text{and} \quad n = 5a + 2b$$

then $\text{hcf}(m, n) = \text{hcf}(a, b)$

Proof.

Let $d = \text{hcf}(a, b)$ and $c = \text{hcf}(m, n)$.

Since d is the highest common factor of a and b , by Definition 4.5, HB p15,

$$d \mid a \text{ and } d \mid b$$

Then by Properties of division (Theorem 4.4, Chapter 1 p15),

$$d \mid (3a + b) \text{ and } d \mid (5a + 2b)$$

Hence,

$$d \mid m \text{ and } d \mid n$$

Therefore, by Definition 4.5,

$$d \mid c$$

And since c is the highest common factor of m and n , by Definition 4.5, HB p15,

$$c \mid m \text{ and } c \mid n$$

Then by Theorem 4.4,

$$c \mid (2m - n) \text{ and } c \mid (5m - 3n)$$

Tell me that $a = 2m - n$ and $b = 5m - 3n$.

Hence,

$$c \mid a \text{ and } c \mid b$$

Therefore, by Definition 4.5, HB p15,

$$c \mid d$$

Thus, $d \mid c$ and $c \mid d \implies c = d$

$$\text{hcf}(m, n) = \text{hcf}(a, b)$$

3/3

□

c)

A number of the form $14a + 3$, where a is a non-negative integer, must have a prime divisor of this same form $14b + 3$, where b is a non-negative integer.

This one is false (as you say below), so why are you writing a "proof" of it?

Proof.

Consider if n is prime, then n is a prime divisor of itself and since $n = 14a + 3$, n is of the form $14b + 3$ where $b = a$.

Now consider if n is composite, then by Fundamental theorem of arithmetic (Theorem 1.7, Chapter 2 HB p17), n can be expressed as;

$$n = p_1^{k_1} p_2^{k_2} \dots p_r^{k_r}$$

thus having at least one prime factor such that $p \mid n$

Hence we will have to show that;

$$n \equiv 3 \pmod{14}$$

and

$$p \mid n, \text{ such that } p \equiv 3 \pmod{14}$$

Consider the counter example where $n = 185$, that is;

$$n = 185 = 5 \times 37$$

$$5 \equiv 5 \pmod{14}$$

and

$$37 \equiv 9 \pmod{14}$$

2/2

As neither of these prime factors are of the form $14b + 3$ the statement is incorrect.

□

d)

If n is divisible by 15 but not divisible by 9 or 25, then $\tau(n)$ is divisible by 4.

Proof.

By Theorem 1.7, HB p17, we can express n as a product of its prime factors;

$$n = 3^{k_1} \cdot 5^{k_2} \cdot p_1^{k_1} \cdot p_2^{k_2} \dots$$

where $k_1, k_2 = 1$ as n is divisible by 15 but not by 9 or 25.

Hence,

$$n = 3^1 \cdot 5^1 \cdot p_1^{k_1} \cdot p_2^{k_2} \dots$$

Then by Proposition 2.3, Chapter 2 p17 (Formula for $\tau(n)$);

$$\begin{aligned} \tau(n) &= (1+1)(1+1)(k_1+1)(k_2+1) \dots \\ &= 2 \cdot 2 \cdot (k_1+1)(k_2+1) \dots \\ &= 4 \cdot (k_1+1)(k_2+1) \dots \end{aligned}$$

great!
3/3

Since $(k_1+1)(k_2+1) \dots$ is an integer, $\tau(n)$ is divisible by 4

□

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Question 5:**a)****i.**

$$3x \equiv 4 \pmod{5}$$

$\text{hcf}(3, 5) = 1$, hence 3 has a multiplicative inverse modulo 5

Hence, this will be the unique solution to the linear congruence

$$3v \equiv 1 \pmod{5}$$

By trying values for v we find that;

$$3 \cdot 2 \equiv 1 \pmod{5}$$

Thus, $v = 2$ is the multiplicative inverse of 3 modulo 5

Multiplying both sides of the original congruence by 2 gives;

$$6x \equiv 8 \pmod{5}$$

Reducing 6 and 8 modulo 5 gives;

$$x \equiv 3 \pmod{5}$$

Thus, the solution is;

$$\boxed{x \equiv 3 \pmod{5}}$$

ii.

$$3x - 1 \equiv 2(4 + x) \pmod{7}$$

distributing the 2 on the RHS;

$$3x - 1 \equiv 8 + 2x \pmod{7}$$

subtracting $2x$ from both sides;

$$x - 1 \equiv 8 \pmod{7}$$

adding 1 to both sides;

$$x \equiv 9 \pmod{7}$$

reducing 9 modulo 7 gives;

$$x \equiv 2 \pmod{7}$$

Thus, the solution is;

$$x \equiv 2 \pmod{7}$$


iii.

$$2(7 - x) \equiv 8 - x \pmod{17}$$

distributing the 2 on the LHS;

$$14 - 2x \equiv 8 - x \pmod{17}$$

adding $2x$ to both sides;

$$14 \equiv 8 + x \pmod{17}$$

subtracting 8 from both sides;

$$6 \equiv x \pmod{17}$$

5/5

Thus, the solution is;

$$x \equiv 6 \pmod{17}$$


b)

$$x \equiv 3 \pmod{5}$$

$$x \equiv 2 \pmod{7}$$

$$x \equiv 6 \pmod{17}$$

$$\text{hcf}(5, 7) = \text{hcf}(5, 17) = \text{hcf}(7, 17) = 1$$

Since 5, 7, and 17 are coprime, by the Chinese Remainder Theorem (CRT, Theorem 4.2, Chapter 3 p20) there is a unique solution modulo $5 \times 7 \times 17 = 595$. Checking shows $x = 23$ satisfies all three congruences, so the least positive solution is;

$$\boxed{x = 23}.$$

I need more working than this to award the method marks - otherwise I don't know that you didn't simply put these equations into a computer to solve.

1/3

6/8

Question 6:**a)**

$$25^{60} \pmod{59}$$

Using Fermat's little theorem (FLT, Theorem 1.1, Chapter 4 p21);

$a^{p-1} \equiv 1 \pmod{p}$ where p is a prime and a is not divisible by p

$$25^{58} \equiv 1 \pmod{59}$$

Substitute this into the original expression, and using index laws;

$$\equiv 25^{58} \cdot 25^2 \pmod{59}$$

Reducing 25^{58} modulo 59 gives;

$$\equiv 1 \cdot 25^2 \pmod{59}$$

$$\equiv 625 \pmod{59}$$

Reducing 625 modulo 59 gives;

$$\equiv 35 \pmod{59}$$

Thus, the solution is;

$$\boxed{35}$$

3/3

b)

$$a^{25} \equiv ? \pmod{195}$$

Proof.

Since $195 = 3 \times 5 \times 13$ and these are coprime, we can use CRT and show;

$$a^{25} \equiv a \pmod{p}$$

For each of the primes divisors $p \in \{3, 5, 13\}$

Using FLT we need to check that $25 \equiv 1 \pmod{p-1}$ for each prime:

I don't understand this notation

$$\begin{array}{l} p = 3 : 25 \equiv 1 \pmod{2} \\ p = 5 : 25 \equiv 1 \pmod{4} \\ p = 13 : 25 \equiv 1 \pmod{12} \end{array} \quad \left. \vphantom{\begin{array}{l} p = 3 : 25 \equiv 1 \pmod{2} \\ p = 5 : 25 \equiv 1 \pmod{4} \\ p = 13 : 25 \equiv 1 \pmod{12} \end{array}} \right\} X$$

Hence, for each prime p dividing 195,

$$a^{25} \equiv a^1 \equiv a \pmod{p}$$

By the CRT, there is a unique solution modulo 195 and so;

$$a^{25} \equiv a \pmod{195}$$

for all integers a .

□

2/4

5/7

Question 7:

$$P(x) = x^3 + 23x^2 + 10x + 6$$

a)

1/1

The degree of $P(x)$ is 3.**b)****i.**

$$P(x) \equiv 7 \pmod{3}$$

$$P(x) \equiv x^3 + 23x^2 + 10x + 6 \pmod{3}$$

$$\equiv x^3 + 2x^2 + x \pmod{3}$$

Factoring out an x gives;

$$\equiv x(x^2 + 2x + 1) \pmod{3}$$

Factor out x and complete the square;

$$\equiv x(x^2 + 2x + 1) \pmod{3}$$

$$\equiv x(x+1)^2 \pmod{3}$$

2/2

Thus the roots satisfy $x \equiv 0$ or $x \equiv -1 \equiv 2 \pmod{3}$. 

$$\boxed{x \equiv 0, 2 \pmod{3}}$$

ii.

$$P(x) \equiv 0 \pmod{7}$$

$$P(x) \equiv x^3 + 23x^2 + 10x + 6 \pmod{7}$$

$$\equiv x^3 + 2x^2 + 3x + 6 \pmod{7}$$

Trying values for x from 0 to 6 gives;

please give details

$$x = 2 \equiv 0 \pmod{7}$$

$$x = 5 \equiv 0 \pmod{7}$$

3/3

Thus, the solutions are;

$$\boxed{x \equiv 2, 5 \pmod{7}}$$



c)

$$P(x) \equiv 0 \pmod{63}$$

Using the CRT we can split this into two congruences;

$$P(x) \equiv 0 \pmod{7}$$

$$P(x) \equiv 0 \pmod{9}$$

$$\text{hcf}(7, 9) = 1$$

$$P(x) \equiv x^3 + 23x^2 + 10x + 6 \pmod{9}$$

$$\equiv x^3 + 5x^2 + x + 6 \pmod{9}$$

Trying values for x from 0 to 8 gives;

Note that you only need to check those values that are also solutions mod 3

$$x = 2 \equiv 0 \pmod{9}$$

$$x = 3 \equiv 0 \pmod{9}$$

$$x = 5 \equiv 0 \pmod{9}$$

$$x = 8 \equiv 0 \pmod{9}$$

Thus, the solutions are;

$$x \equiv 2, 3, 5, 8 \pmod{9}$$

Using the solutions from part (b) we have;

$$x \equiv 2, 5 \pmod{7}$$

$$x \equiv 2, 3, 5, 8 \pmod{9}$$

$$\text{hcf}(7, 9) = 1$$

Since 7 and 9 are coprime, by the CRT there is a unique solution modulo $7 \times 9 = 63$.

first pair

$$x \equiv 2 \pmod{7}$$

$$x \equiv 2 \pmod{9}$$

$$\implies x \equiv 2 \pmod{63}$$

Now using the multiplicative inverse v of 7 (mod 9);

$$v = 4, \text{ since } 7 \times 4 \equiv 1 \pmod{9}$$

second pair

$$x \equiv 2 \pmod{7}$$

$$x \equiv 3 \pmod{9}$$

$$x = 7k + 2$$

$$7k + 2 \equiv 3 \pmod{9}$$

Substituting v and multiplying both sides by 4 gives;

$$4(7k + 2) \equiv 4 \cdot 3 \pmod{9}$$

$$k + 8 \equiv 12 \pmod{9}$$

$$k \equiv 4 \pmod{9}$$

Substituting back gives;

$$x = 7(9m + 4) + 2$$

$$= 63m + 30$$

$$\Rightarrow x \equiv 30 \pmod{63}$$



Checking the rest of the pairs, shows $x = 2, 5, 16, 23, 30, 37, 47, 54$ satisfy both congruences, so the least positive solutions are;

2, 5, 16, 23, 30, 37, 47, 54

3/4

You should get 2,5,12,23,26,30,44,47

9/10

Question 8:**a)**

I have watched the online session entitled "M303: Things you need to know".

b)

PA **Paul Allen**
4 Oct 2025 21:32

Has anybody found anything about what we should be calling highest common factor in our proofs? I typeset everything in LaTeX and it has `\gcd` for use in proofs of ratios very reason, but our text constantly refers to it as highest common factor. I don't mind declaring a new operator in my `.sty` file, but I feel like if `\gcd` is there, it's there because that is the industry standard.

Has anybody come across this or similar typesetting issues?

Thanks

Paul

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<https://learn2.open.ac.uk/mod/forumng/discuss.php?d=5080556>

3/3

Part B

Question 9: Question 10 from the TMA booklet

a)

For $k \geq 1$ let $P(n)$ be the statement;

"Let k be the number of blue eared aliens, all the aliens will leave at 6am on the k^{th} day."

This is part of the answer to (b). For (a) you should just state what you are aiming to prove in (b)

b)

Base case:

$$P(1) : k = 1$$

If a blue eared alien does not see any other blue eared aliens, then they will know they are the only blue eared alien on the planetoid, and will leave at 6am on the next rotation, i.e the k^{th} day.

So $P(1)$ holds

Inductive step: Assume that the statement is true for some arbitrary $m \geq 1$, i.e if there are m blue eared aliens, they will all leave at 6am on the m^{th} day.

Now consider if there are $m + 1$ blue eared aliens. Each blue eared alien will see m other blue eared aliens. When on the m^{th} day no aliens leave, they will realise that they must also have blue ears, so all $m + 1 = k$ aliens will leave at 6am on the k^{th} day.

Why is this true? Refer to your induction hypothesis

Thus, by the principle of mathematical induction, the statement is true for all $k \geq 1$.

v good

7/8

9/10

Question 10: Question 12 from the TMA booklet

Let p and q be distinct odd primes.

a)

An integer x satisfied $x^2 \equiv 1 \pmod{pq}$ if, and only if, x satisfies both

$$x^2 \equiv 1 \pmod{p} \text{ and } x^2 \equiv 1 \pmod{q}$$

Forward direction ($\Rightarrow$):

$$x^2 \equiv 1 \pmod{pq}$$

$$x^2 - 1 \equiv 0 \pmod{pq}$$

$$(x-1)(x+1) \equiv 0 \pmod{pq} \quad \text{No need to factorize in your argument}$$

Hence, $pq \mid (x-1)(x+1)$ Since p and q are distinct primes;

$$p \mid (x-1)(x+1) \text{ and } q \mid (x-1)(x+1)$$

Thus,

$$x^2 \equiv 1 \pmod{p} \text{ and } x^2 \equiv 1 \pmod{q}$$



Backward direction ($\Leftarrow$):

$$x^2 \equiv 1 \pmod{p} \text{ and } x^2 \equiv 1 \pmod{q}$$

$$x^2 - 1 \equiv 0 \pmod{p} \text{ and } x^2 - 1 \equiv 0 \pmod{q}$$

$$(x-1)(x+1) \equiv 0 \pmod{p} \text{ and } (x-1)(x+1) \equiv 0 \pmod{q}$$

Hence,

$$p \mid (x-1)(x+1) \text{ and } q \mid (x-1)(x+1)$$

Since p and q are distinct primes;

$$pq \mid (x-1)(x+1)$$

$$(x-1)(x+1) \equiv 0 \pmod{pq}$$

$$x^2 - 1 \equiv 0 \pmod{pq}$$

$$x^2 \equiv 1 \pmod{pq}$$

Therefore,

$$x^2 \equiv 1 \pmod{pq} \iff x^2 \equiv 1 \pmod{p} \text{ and } x^2 \equiv 1 \pmod{q}$$

2/2

b)

$$x^2 \equiv 1 \pmod{p}$$

As p is odd, $p-1$ is even, hence divisible by 2.

By Proposition 4.8, Chapter 4 p22, if p is prime and d is a divisor of $p-1$ then the congruence $x^d - 1 \equiv 0 \pmod{p}$ has exactly d solutions.

Thus, $x^2 \equiv 1 \pmod{p}$ has exactly 2 solutions. These solutions are $x \equiv 1 \pmod{p}$ and $x \equiv -1 \equiv p-1 \pmod{p}$.

You need to mention that 1 and -1 are solutions, and that they are *distinct* because p is odd and hence not 2.

2/2

c)

$$x^2 \equiv 1 \pmod{pq}$$

$$x^2 \equiv 1 \pmod{p}$$

$$x^2 \equiv 1 \pmod{q}$$

$$x^2 \equiv 1 \pmod{p}$$

$$x^2 - 1 \equiv 0 \pmod{p}$$

$$(x-1)(x+1) \equiv 0 \pmod{p}$$

Hence, $p \mid (x-1)$ or $p \mid (x+1)$

Thus, the solutions are;

$$x \equiv 1 \pmod{p} \text{ or } x \equiv -1 \equiv p-1 \pmod{p}$$

The same argument can be used for q . Hence,

$$x^2 \equiv 1 \pmod{pq} \text{ has four solutions.}$$

You haven't used the Chinese remainder theorem (which is asked). What you have here is only part of the argument.

2/4

d)

$$x^2 \equiv 1 \pmod{95}$$

Using the CRT we can split this into two congruences;

$$x^2 \equiv 1 \pmod{5}$$

$$x^2 \equiv 1 \pmod{19}$$

$$\text{hcf}(5, 19) = 1$$

Since 5 and 19 are coprime, by the CRT there is a unique solution modulo $5 \times 19 = 95$.

The solutions to $x^2 \equiv 1 \pmod{5}$ are;

$$x \equiv 1 \pmod{5} \text{ or } x \equiv 4 \pmod{5} \quad \checkmark$$

The solutions to $x^2 \equiv 1 \pmod{19}$ are;

$$x \equiv 1 \pmod{19} \text{ or } x \equiv 18 \pmod{19} \quad \checkmark$$

19 is 0 (mod 19), so clearly cannot solve the mod 19 congruences.

Checking the four pairs of solutions shows $x = 1, 4, 16, 19$ satisfy both congruences, so the least positive solutions are;

1/3

1, 4, 16, 19

$\times$ $1, 39, 56, 94 \equiv -1 \pmod{95}$

e)

This does not contradict Lagrange's theorem (Theorem 4.3, Chapter 4 p22) as this theorem only applies to polynomials modulo a prime number, and 95 is not prime.

2/2

9/13